

S E PAVITHRA

Chennai, India • +91-8925298656 • spavithraaug8@gmail.com • linkedin.com/in/pavithra-cse • github.com/pavithra123456789

PROFESSIONAL SUMMARY

Final-year Computer Science undergraduate at HITS Chennai (CGPA: 9.40/10) with hands-on experience in Python, NLP pipelines, LLM-based agentic workflows, and AI application development. Built real-world AI tools including a multi-stage NLP resume analysis pipeline and a multi-step agentic image generation system using Hugging Face APIs and prompt engineering. Familiar with RAG concepts, LangChain fundamentals, REST APIs, and Git. Passionate about AI, machine learning, and building intelligent automated systems.

TECHNICAL SKILLS

Languages: Python (Primary), Java, JavaScript, SQL

AI / ML & LLMs: NLP Pipelines, TF-IDF, Machine Learning Fundamentals, Hugging Face APIs, LangChain (Learning), LlamaIndex (Learning), RAG Concepts, Prompt Engineering, LLM Concepts

Data & Libraries: Pandas, NumPy, Data Preprocessing, Model Evaluation

Backend & APIs: REST APIs, Flask, Spring Boot, JWT Authentication

Tools: Git, GitHub, Docker, Postman, Jupyter Notebooks, Linux

PROFESSIONAL EXPERIENCE

Data Integration Intern | NSIIC (Chennai)

Jul 2023

- Collected, cleaned, and preprocessed large structured datasets using Python (Pandas/NumPy) and SQL for automated pipelines.
- Performed data validation and quality checks, maintaining high consistency and accuracy for downstream analytical use.
- Documented data methodologies and pipeline processes clearly for reproducibility and team knowledge sharing.

Full Stack Developer Intern | Codebind Technologies (Chennai)

Feb 2025 – May 2025

- Developed RESTful backend services and web applications using Python-based frameworks and SQL databases.
- Implemented JWT-based authentication and secure backend pipelines for multi-role user session management.
- Debugged critical application defects through systematic root-cause analysis, improving reliability and performance.

Cybersecurity Intern | NovaNectar Services Pvt. Ltd. (Remote)

Jun 2024

- Developed Python automation scripts for log analysis, identifying patterns and anomalies within large system datasets.
- Collaborated with cross-functional teams in an agile remote environment, presenting findings clearly to stakeholders.
- Performed systematic assessments and contributed to technical documentation and process improvement.

PROJECTS

AI Student Developer Kit | Python, NLP, TF-IDF, Hugging Face APIs, REST APIs, LangChain Concepts

- Designed a multi-stage agentic AI pipeline chaining resume parsing, NLP skill-gap analysis, and recommendation generation using Python.
- Applied TF-IDF, NLP techniques, and Hugging Face APIs to evaluate and iterate on model outputs across 100+ varied inputs.
- Documented experiments, evaluation metrics, and pipeline improvements for reproducibility and iterative development.

AI Image Generator | Python, Hugging Face APIs, Prompt Engineering, Agentic Workflow

- Built a multi-step agentic workflow with prompt chaining, retry logic, and structured automated evaluation across 200+ test runs.
- Applied few-shot and chain-of-thought prompt engineering to fine-tune LLM outputs and improve generation alignment.
- Designed plug-and-play modular architecture enabling rapid model swaps — demonstrating scalable agentic AI design.

TaskFlow API — Task Management System | Java, Spring Boot, MySQL, JWT, REST APIs

- Built a RESTful backend API with CRUD endpoints, JWT authentication, and role-based access control using Spring Boot and MySQL.
- Containerized using Docker and validated all API workflows via Postman across deployment environments.
- Designed layered architecture with clean code principles ensuring maintainability and scalability.

EDUCATION & CERTIFICATIONS

B.Tech – Computer Science Engineering | Hindustan Institute of Technology and Science, Chennai

CGPA: 9.40 | 2022–2026

Certifications: Oracle Cloud Infra. 2025 AI Foundations Associate • Python for Everyone (Udemy) • Pen Testing (Udemy)